

Chrome Dino Pure Software Raster Engine

Complete Line-by-Line Code Dissection, Mathematical Derivations & Examiner Viva Defense Manual

PURE CPU RASTERIZER

7 CORE CG ALGORITHMS

ZERO GPU CALLS

LINE-BY-LINE ANNOTATIONS

1. System Architecture: The Journey of a Frame (RAM to Screen)

In modern computer graphics, high-level frameworks (OpenGL, DirectX, Vulkan, WebGL) delegate line drawing, circle math, polygon filling, and clipping to dedicated GPU hardware shader pipelines. In contrast, **this engine implements every single operation from first mathematical principles on the CPU.**

The Three Pillars of Our Pure Software Raster Pipeline

- Contiguous Memory Model:** All 240,000 pixels (800 × 300) reside in an uninterrupted 1D array of 32-bit integers in system RAM (`std::vector<uint32_t>`).
- Algorithmic Geometry Synthesis:** Lines, circles, ellipses, polygons, and text are converted into discrete coordinate indices using integer-only mathematical decision variables.
- Zero-Copy Single Blit Presentation:** At the end of every frame (60 FPS), the completed pixel memory buffer is wrapped by a `QImage` pointer without copying and blitted to the monitor in one atomic paint event.

1.1 Memory Indexing & Flat 1D RAM Mapping

Screens are 2-dimensional Cartesian grids with coordinates (x, y), where (0, 0) is the top-left corner, x increases horizontally to the right (0 ≤ x < 800), and y increases vertically downward (0 ≤ y < 300).

```
// 2D Coordinate to 1D RAM Linear Memory Index Formula:
size_t memoryIndex = (y * width) + x;
uint32_t pixelValue = m_pixels[memoryIndex]; // 0xAARRGGBB in physical RAM
```

Design Choice	How It Works in Machine Memory	Why We Did This (Viva Explanation)
1D Flat Array vs 2D Array	<code>vector<uint32_t></code> (960 KB contiguous block) instead of <code>uint32_t**</code> (array of pointers).	Eliminates pointer indirection: An array of row pointers causes 300 cache misses per frame. A contiguous 1D array allows the CPU L1/L2 data cache to prefetch entire raster scanlines sequentially.
32-Bit Aligned ARGB (uint32_t)	Each pixel is exactly 4 bytes: Alpha (bits 24-31), Red (bits 16-23), Green (bits 8-15), Blue (bits 0-7).	Single-cycle 32-bit ALU word alignment: Modern 64-bit and 32-bit CPUs can write a 4-byte pixel in a single machine instruction. 24-bit RGB causes unaligned memory penalties.
Bitwise Color Packing	<code>(A << 24) (R << 16) (G << 8) B</code>	Bit shifts and bitwise OR take exactly 1 clock cycle on the CPU ALU, completely bypassing slow division or floating-point multiplication.

2. Direct Framebuffer Memory Engine (Framebuffer.h / .cpp)

The `Framebuffer` class represents physical video memory (VRAM) simulated entirely in host RAM.

2.1 Framebuffer::setPixelBlend (Alpha Compositing Engine)

```
1 void Framebuffer::setPixelBlend(int x, int y, uint32_t srcColor) {
2     if (x < 0 || x >= m_width || y < 0 || y >= m_height) return;
3     uint8_t sa = (srcColor >> 24) & 0xFF;
4     if (sa == 0) return;
5     if (sa == 255) { m_pixels[y * m_width + x] = srcColor; m_pixelsWritten++; return; }
6     uint32_t dstColor = m_pixels[y * m_width + x];
7     uint8_t sr = (srcColor >> 16) & 0xFF, sg = (srcColor >> 8) & 0xFF, sb = srcColor & 0xFF;
8     uint8_t dr = (dstColor >> 16) & 0xFF, dg = (dstColor >> 8) & 0xFF, db = dstColor & 0xFF;
9     uint32_t invA = 255 - sa;
10    uint8_t outR = static_cast<uint8_t>((sr * sa + dr * invA) / 255);
11    uint8_t outG = static_cast<uint8_t>((sg * sa + dg * invA) / 255);
12    uint8_t outB = static_cast<uint8_t>((sb * sa + db * invA) / 255);
13    m_pixels[y * m_width + x] = packARGB(255, outR, outG, outB);
14    m_pixelsWritten++;
15 }
```

Line #	C++ Code Snippet	Plain Language Explanation	Why Written This Way & Examiner Defense
Line 2	if (x < 0 x >= m_width ...) return;	Guards against coordinates falling outside viewport.	Memory Safety: Prevents buffer overflow. Writing to negative indices corrupts heap headers and causes an instant segmentation fault (SIGSEGV).
Line 3	uint8_t sa = (srcColor >> 24) & 0xFF;	Extracts the high 8 bits of the 32-bit color word.	ARGB byte order has Alpha at bits 24-31. Bit shifting right by 24 and masking with <code>0xFF</code> isolates alpha (0 ... 255).
Line 4-5	if (sa == 0) return; if (sa == 255) ...	Fast-path optimization for transparent or opaque pixels.	Speed Optimization: If alpha is 0, skip all math. If 255, directly overwrite destination. Avoids 6 multiplications and 3 divisions for 90% of game pixels!
Line 7-8	uint8_t sr = (srcColor >> 16) & 0xFF...	Unpacks Red, Green, and Blue channels for source and destination.	Separates composite integer into distinct color colorimetric components for linear color channel blending.
Line 9	uint32_t invA = 255 - sa;	Calculates destination weighting factor ($1 - \alpha$).	Implements standard Porter-Duff <i>Source-Over</i> compositing equation: $C_{out} = C_{src}\alpha + C_{dst}(1 - \alpha)$.
Line 10-12	(sr * sa + dr * invA) / 255	Computes weighted average of colors on integer ALU.	Pure integer arithmetic. The maximum product is $255 \times 255 + 255 \times 255 = 130,050$, which fits comfortably inside a standard 32-bit unsigned integer without overflow.
Line 13	m_pixels[...] = packARGB(...)	Repacks components and writes to target RAM address.	Stores the finalized blended pixel back into the flat 1D memory array in 1 machine store instruction.

3. Algorithm 1: Bresenham's Integer Line Algorithm

Implemented in `SoftwareRasterizer.cpp:31-55` . Generates continuous pixel lines between endpoints (x_0, y_0) and (x_1, y_1) without floating point calculations, rounding functions, or division.

```
1 void SoftwareRasterizer::drawLineBresenham(int x0, int y0, int x1, int y1, uint32_t color) {
2     int dx = std::abs(x1 - x0);
3     int dy = std::abs(y1 - y0);
4     int sx = (x0 < x1) ? 1 : -1;
5     int sy = (y0 < y1) ? 1 : -1;
6     int err = dx - dy;
7
8     while (true) {
9         setPixel(x0, y0, color);
10        if (x0 == x1 && y0 == y1) break;
11        int e2 = 2 * err;
12        if (e2 > -dy) {
13            err -= dy;
14            x0 += sx;
15        }
16        if (e2 < dx) {
17            err += dx;
18            y0 += sy;
19        }
20    }
21 }
```

Line #	C++ Code Snippet	Plain Language Explanation	Mathematical Rationale & Viva Trap Questions
Line 2-3	<code>int dx = abs(x1 - x0);</code> <code>int dy = abs(y1 - y0);</code>	Computes total spans along horizontal and vertical axes.	Absolute differences ensure distances are non-negative, allowing universal treatment regardless of slope direction.
Line 4-5	<code>int sx = (x0 < x1) ? 1 : -1;</code> <code>int sy = (y0 < y1) ? 1 : -1;</code>	Determines the sign of movement along each axis.	All 8 Octants Coverage: Enables the algorithm to draw lines in any direction (left-to-right, right-to-left, top-to-bottom, or bottom-to-top) without branching into 8 separate routines.
Line 6	<code>int err = dx - dy;</code>	Initializes the integer decision variable.	Why <code>dx - dy</code> ? Balances the horizontal and vertical error terms in a single variable. When $e_2 > -dy$, x must step; when $e_2 < dx$, y must step.
Line 9	<code>setPixel(x0, y0, color);</code>	Plots current discrete coordinate into the Framebuffer.	Writes directly to RAM with automatic bounds-clipping protection.
Line 10	<code>if (x0 == x1 && y0 == y1) break;</code>	Loop termination guard.	Breaks immediately when target destination coordinate is reached, preventing infinite looping or overshoot.
Line 11	<code>int e2 = 2 * err;</code>	Scales error by 2 into temporary snapshot variable <code>e2</code> .	Critical Viva Question: "Why do you store <code>2 * err</code> in <code>e2</code> ?" Answer: Line 13 modifies <code>err</code> . If we checked <code>2 * err</code> directly in Line 16, it would read the already-mutated error and skip the vertical step on diagonal lines!
Line 12-15	<code>if (e2 > -dy) { err -= dy; x0 += sx; }</code>	Evaluates whether the ideal line crossed half a pixel along X .	If true, advances x_0 by step direction s_x and adjusts error accumulator by subtracting vertical delta dy .

Line #	C++ Code Snippet	Plain Language Explanation	Mathematical Rationale & Viva Trap Questions
Line 16-19	<pre>if (e2 < dx) { err += dx; y0 += sy; }</pre>	Evaluates whether the ideal line crossed half a pixel along Y.	If true, advances y_0 by step direction s_y and increments error accumulator by horizontal delta dx . On a 45° line, both conditions trigger simultaneously (diagonal step).

4. Algorithm 2: Midpoint Circle Algorithm (Bresenham's Circle)

Implemented in `SoftwareRasterizer.cpp:195-230`. Evaluates the implicit circle equation $F(x, y) = x^2 + y^2 - R^2 = 0$. Uses **8-way symmetry** to calculate only one 45° octant, plotting 8 pixels per mathematical evaluation.

```
1 void SoftwareRasterizer::drawCircleMidpoint(int xc, int yc, int r, uint32_t color) {
2     if (r <= 0) { setPixel(xc, yc, color); return; }
3     int x = 0;
4     int y = r;
5     int d = 1 - r; // Initial decision parameter (Midpoint test)
6
7     auto plot8 = [&](int px, int py) {
8         setPixel(xc + px, yc + py, color); setPixel(xc - px, yc + py, color);
9         setPixel(xc + px, yc - py, color); setPixel(xc - px, yc - py, color);
10        setPixel(xc + py, yc + px, color); setPixel(xc - py, yc + px, color);
11        setPixel(xc + py, yc - px, color); setPixel(xc - py, yc - px, color);
12    };
13    plot8(x, y);
14    while (x < y) {
15        x++;
16        if (d < 0) {
17            d += 2 * x + 1;
18        } else {
19            y--;
20            d += 2 * (x - y) + 1;
21        }
22        plot8(x, y);
23    }
24 }
```

Line #	C++ Code Snippet	Plain Language Explanation	Mathematical Rationale & Viva Trap Questions
Line 3-4	int x = 0; int y = r;	Starts tracking at the top apex of the circle (0, R).	At (0, R), the slope of the circle tangent is 0 (horizontal). Stepping in x ensures a continuous curve without gaps.
Line 5	int d = 1 - r;	Calculates initial decision parameter.	Textbook Derivation: Evaluating midpoint (1, R - 0.5) gives $F(1, R - 0.5) = 1 + (R^2 - R + 0.25) - R^2 = \frac{5}{4} - R$. Since R is an integer, $\frac{5}{4} - R$ has the exact same sign as $1 - R$ for all integer steps, saving float operations!
Line 7-12	plot8(px, py)	Lambda generating all 8 symmetric octants simultaneously.	8-Way Symmetry: Circle has symmetry across X-axis, Y-axis, and diagonal line $y = x$. Computing 1 octant (0° ... 45°) produces all 8 octants, saving 87.5% CPU work!
Line 14	while (x < y)	Loop condition running from 0° up to 45° diagonal.	At $x = y$, the tangent slope reaches -1. Beyond this point, x and y swap roles, which is already handled by 8-way symmetry.
Line 15	x++;	Always advances 1 pixel along the horizontal axis.	In Octant 2 (45° ... 90°), $ dx > dy $, making X the primary driver axis.
Line 16-17	if (d < 0) { d += 2*x + 1; }	Midpoint is inside circle: Choose East pixel (x+1, y).	y remains unchanged. Decision increment is $\Delta E = 2x_{k+1} + 1$.
Line 18-21	else { y--; d += 2*(x - y) + 1; }	Midpoint is outside circle: Choose South-East pixel (x+1, y-1).	Decrements y. Decision increment is $\Delta SE = 2(x_{k+1} - y_{k+1}) + 1$.

5. Algorithm 3: Midpoint Ellipse Algorithm

Implemented in `SoftwareRasterizer.cpp:277-331`. Evaluates implicit ellipse equation $b^2x^2 + a^2y^2 - a^2b^2 = 0$. Uses **4-way quadrant symmetry** and divides the curve into **Two Distinct Slope Regions**.

```
1 void SoftwareRasterizer::drawEllipseMidpoint(int xc, int yc, int rx, int ry, uint32_t color) {
2     long long a = rx, b = ry, a2 = a * a, b2 = b * b;
3     long long x = 0, y = b;
4     // REGION 1: |Slope| < 1 (x is primary driver)
5     double p1 = b2 - (a2 * b) + (0.25 * a2);
6     long long dx = 2 * b2 * x, dy = 2 * a2 * y;
7     while (dx < dy) {
8         plot4(x, y); x++; dx += 2 * b2;
9         if (p1 < 0) p1 += dx + b2;
10        else { y--; dy -= 2 * a2; p1 += dx - dy + b2; }
11    }
12    // REGION 2: |Slope| >= 1 (y is primary driver)
13    double p2 = b2*(x+0.5)*(x+0.5) + a2*(y-1)*(y-1) - a2*b2;
14    while (y >= 0) {
15        plot4(x, y); y--; dy -= 2 * a2;
16        if (p2 > 0) p2 += a2 - dy;
17        else { x++; dx += 2 * b2; p2 += dx - dy + a2; }
18    }
19 }
```

Line #	C++ Code Snippet	Plain Language Explanation	Mathematical Rationale & Viva Trap Questions
Line 2	long long a = rx, b = ry...	Uses 64-bit integers for squared radius terms.	Overflow Protection: $a^2 \times b^2$ can easily exceed 2×10^9 for radii above 250 pixels, overflowing standard 32-bit signed integers. <code>long long</code> prevents arithmetic wrap-around.
Line 5-7	double p1 = b2 - (a2 * b) + ... while (dx < dy)	Region 1: Moves while curve slope magnitude $ dy/dx < 1$.	Why Two Regions? The slope of an ellipse changes from 0 to $-\infty$. In Region 1, slope is flat, so we step along X. The boundary occurs exactly where $2b^2x = 2a^2y$.
Line 8-10	if (p1 < 0) p1 += dx + b2; else { y--; ... }	Region 1 decision parameter update.	Tests midpoint $(x+1, y-0.5)$. If $p_1 < 0$, midpoint is inside: choose East. If $p_1 \geq 0$, choose South-East.
Line 13-14	double p2 = ... while (y >= 0)	Region 2: Slope is steep ($ dy/dx \geq 1$), stepping along Y.	Stepping in X here would create jagged gaps in the vertical sides. Stepping in Y guarantees contiguous pixels until $y = 0$.
Line 16-17	if (p2 > 0) p2 += a2 - dy; else { x++; ... }	Region 2 decision parameter update.	Tests midpoint $(x+0.5, y-1)$. If $p_2 > 0$, choose South. If $p_2 \leq 0$, choose South-East.

6. Algorithm 4: Scanline Polygon Fill (Parity Fill)

Implemented in `SoftwareRasterizer.cpp:410-451`. Fills arbitrary convex, concave, or self-intersecting polygons using the **Even-Odd Parity Rule** and edge linear interpolation.

```
1 void SoftwareRasterizer::fillPolygonScanline(const std::vector<QPoint>& vertices, uint32_t color) {
2     if (vertices.size() < 3) return;
3     int minY = m_fb.height(), maxY = 0;
4     for (const auto& pt : vertices) { minY = std::min(minY, pt.y()); maxY = std::max(maxY, pt.y()); }
5     minY = std::max(0, minY); maxY = std::min(m_fb.height() - 1, maxY);
6     std::vector<int> nodeX;
7
8     for (int y = minY; y <= maxY; ++y) {
9         nodeX.clear();
10        for (size_t i = 0; i < vertices.size(); ++i) {
11            size_t next = (i + 1) % vertices.size();
12            int y0 = vertices[i].y(), y1 = vertices[next].y();
13            int x0 = vertices[i].x(), x1 = vertices[next].x();
14            if ((y0 < y && y1 >= y) || (y1 < y && y0 >= y)) {
15                int intersectX = x0 + static_cast<int>(std::round(((double)(y - y0)/(y1 - y0))*(x1 - x0)));
16                nodeX.push_back(intersectX);
17            }
18        }
19        std::sort(nodeX.begin(), nodeX.end());
20        for (size_t i = 0; i + 1 < nodeX.size(); i += 2) {
21            drawHorizontalSpan(nodeX[i], nodeX[i + 1], y, color);
22        }
23    }
24 }
```

Line #	C++ Code Snippet	Plain Language Explanation	Mathematical Rationale & Viva Trap Questions
Line 3-5	<code>minY = min(minY, pt.y())...</code>	Finds vertical bounding extent $[y_{\min}, y_{\max}]$.	Restricts scanline testing to only rows intersecting the polygon bounding box, skipping empty rows.
Line 11	<code>size_t next = (i + 1) % n;</code>	Wraps the last vertex back to vertex 0.	Guarantees that the polygon boundary is a closed geometric loop.
Line 14	<code>if ((y0 < y && y1 >= y) ...)</code>	Tests if current scanline y straddles polygon edge $(i, next)$.	Asymmetric Inequality Rule: Uses <code><</code> on one vertex and <code>>=</code> on the other. This prevents vertices that sit directly on scanline y from being counted twice, which would corrupt parity!
Line 15	<code>x0 + round(((y - y0)/(y1 - y0)) * (x1 - x0))</code>	Linear interpolation of edge intersection coordinate.	Solves line equation $x = x_0 + ((y - y_0)/(y_1 - y_0)) \cdot (x_1 - x_0)$ at scanline altitude y .
Line 19	<code>std::sort(nodeX.begin(), nodeX.end());</code>	Sorts x-intersection points from left to right.	Enforces ordering so consecutive pairs $(x_0, x_1), (x_2, x_3)$ define the interior spans of the polygon.
Line 20-22	<code>for (size_t i = 0; i+1 < size(); i += 2)</code>	Fills horizontal span between odd and even intersection pairs.	Even-Odd Parity Rule: Ray casting from $-\infty$ flips inside/outside state on each edge intersection: Span $[x_0, x_1]$ is INSIDE, $[x_1, x_2]$ is OUTSIDE, $[x_2, x_3]$ is INSIDE.

7. Algorithm 5 & 6: Affine Rotation & Xiaolin Wu Anti-Aliasing

7.1 Affine Sprite Rotation via Inverse Mapping (SoftwareRasterizer.cpp:573-631)

```
// Inverse Affine Coordinate Transformation Equation:
float srcX = dx * std::cos(rad) - dy * std::sin(rad) + srcCenterX;
float srcY = dx * std::sin(rad) + dy * std::cos(rad) + srcCenterY;
int sx = static_cast<int>(std::round(srcX));
int sy = static_cast<int>(std::round(srcY));
```

Transformation Method	How It Works	Result & Viva Defense
Forward Mapping (Naive)	Iterates through source sprite pixels (u, v) and calculates destination position (x', y') = R · (u, v).	Failure: Due to rounding to integer pixels, multiple source pixels land on the same target pixel, while other target pixels are never hit, creating <i>unpainted black holes</i> .
Inverse Mapping (Our Engine)	Iterates through every destination pixel (x, y) in the bounding circle and pulls the color from source (u, v) = R ⁻¹ · (x, y).	Guaranteed Continuity: Every destination pixel in the target rectangle is evaluated exactly once. Zero holes, zero tears, 100% solid rotation.

7.2 Xiaolin Wu's Anti-Aliased Line Algorithm (SoftwareRasterizer.cpp:109-174)

```
1 for (int x = xpxl1 + 1; x < xpxl2; ++x) {
2     int ipart = static_cast<int>(std::floor(intery));
3     float fpart = intery - ipart;
4     plotWu(x, ipart, 1.0f - fpart); // Upper pixel (main body)
5     plotWu(x, ipart + 1, fpart);   // Lower pixel (anti-aliasing fringe)
6     intery += gradient;
7 }
```

Line #	C++ Code Snippet	Plain Language Explanation	Mathematical Rationale & Sub-Pixel Filtering
Line 2-3	int ipart = floor(intery); float fpart = intery - ipart;	Decomposes real y-intercept into integer and fractional parts.	<code>ipart</code> represents the base pixel row. <code>fpart</code> represents how close the ideal mathematical line is to the next row below (0.0 ... 1.0).
Line 4-5	plotWu(x, ipart, 1.0f - fpart); plotWu(x, ipart+1, fpart);	Plots two vertically adjacent pixels with complementary alpha weights.	Energy Conservation: (1 - <i>fpart</i>) + <i>fpart</i> = 1.0. The total luminous intensity is conserved across the two pixels. The human eye blends them into a smooth vector edge.

8. Game Physics & State Machine Dissection (Dino.cpp)

The Dino is governed by an explicit finite state machine (Idle , Running , Jumping , Ducking , Dead) and Euler numerical integration.

```
1 void Dino::update(float dt, ParticleSystem& particles) {
2     if (m_state == DinoState::Dead) return;
3     if (m_jumpBufferTimer > 0.0f) m_jumpBufferTimer -= dt;
4     if (m_state == DinoState::Running || m_state == DinoState::Ducking) {
5         m_coyoteTimer = 0.08f; // 80ms grounded grace window
6         m_animTimer += dt;
7         if (m_animTimer >= 0.10f) { m_animTimer = 0.0f; m_animFrame = (m_animFrame + 1) % 2; }
8     } else if (m_coyoteTimer > 0.0f) { m_coyoteTimer -= dt; }
9
10    if (m_state == DinoState::Jumping) {
11        float effectiveGravity = (m_isDuckingInput && m_vy > 0.0f) ? m_fastFallGravity : m_gravity;
12        m_vy += effectiveGravity * dt; // v = v0 + a * dt
13        m_y += m_vy * dt; // y = y0 + v * dt
14        float groundY = m_isDuckingInput ? (m_groundY - duckHeight) : (m_groundY - standHeight);
15        if (m_y >= groundY) {
16            m_y = groundY; m_vy = 0.0f;
17            m_state = m_isDuckingInput ? DinoState::Ducking : DinoState::Running;
18            if (m_jumpBufferTimer > 0.0f) { m_jumpBufferTimer = 0.0f; startJump(); }
19        }
20    }
21 }
```

Line #	C++ Code Snippet	Plain Language Explanation	Game Physics Mechanics & Responsive Game Feel
Line 5	m_coyoteTimer = 0.08f;	Refreshes Coyote Time window while feet touch ground.	Coyote Time: Grants an 80ms grace window after walking off an edge or dropping, allowing late jump presses to still trigger. Prevents "eaten input" frustration.
Line 11	effectiveGravity = (ducking && vy > 0) ? fastFall : grav	Fast-fall downward acceleration boost.	If player holds Down Arrow / Duck while falling, gravity doubles (3600 px/s ²). Gives competitive players instant downward recovery over birds.
Line 12-13	m_vy += effectiveGravity * dt; m_y += m_vy * dt;	Euler Numerical Physics Integration.	Updates velocity by acceleration, and position by velocity, scaled by delta-time <i>dt</i> . Ensures physics simulation runs at identical speed on 60 Hz, 144 Hz, or mobile screens.
Line 15-16	if (m_y >= groundY) { m_y = groundY; m_vy = 0; }	Ground collision clamping.	Snaps position precisely to floor level and resets vertical velocity to zero, preventing sub-surface penetration.
Line 18	if (m_jumpBufferTimer > 0.0f) startJump();	Executes buffered jump input immediately on touchdown.	Jump Buffering: If player taps Space up to 140ms before landing, the input is remembered and fired the microsecond the Dino touches ground.

9. Two-Tier Collision Pipeline: AABB Broadphase to Pixel Narrowphase

Implemented in `DinoGame.cpp:340-398` . Avoids $O(N \times W \times H)$ pixel comparisons by filtering geometry through a fast 2-tier detection pipeline.

```
// PHASE 1: BROADPHASE AABB (Axis-Aligned Bounding Box Intersection)
bool AABB::intersects(const AABB& o) const {
    return (x < o.x + o.w && x + w > o.x &&
            y < o.y + o.h && y + h > o.y);
}

// PHASE 2: NARROWPHASE PIXEL-PERFECT BITMASK CHECK
bool checkPixelCollision(const Sprite& s1, int x1, int y1, const Sprite& s2, int x2, int y2, int& hitX, int&
    int left  = std::max(x1, x2),          right  = std::min(x1 + s1.width(), x2 + s2.width());
    int top   = std::max(y1, y2),          bottom = std::min(y1 + s1.height(), y2 + s2.height());
    if (left >= right || top >= bottom) return false;

    for (int y = top; y < bottom; ++y) {
        for (int x = left; x < right; ++x) {
            uint32_t p1 = s1.getPixel(x - x1, y - y1);
            uint32_t p2 = s2.getPixel(x - x2, y - y2);
            if ((p1 >> 24) > 0 && (p2 >> 24) > 0) {
                hitX = x; hitY = y;
                return true; // Exact non-transparent pixel overlap confirmed!
            }
        }
    }
    return false;
}
```

Collision Stage	Mathematical Complexity	Function in Engine & Examiner Defense
Phase 1: Multi-Box AABB Broadphase	$O(1)$ constant time (4 integer comparisons).	Dino uses 3 compound sub-hitboxes (Head, Torso, Feet) while standing and 1 elongated box while ducking. If none intersect the obstacle's AABB, the entire obstacle is discarded in 0.0001 ms.
Phase 2: Raster Mask Narrowphase	$O(\text{overlap area})$ (only the intersection rectangle).	Only executes when bounding boxes collide. Inspects whether <i>both</i> the Dino sprite and Obstacle sprite have non-transparent alpha ($\alpha > 0$) at the exact same world coordinate (x, y). Eliminates "ghost hits" on empty transparent sprite borders!

10. The Master Viva Defense Cheat Sheet (Top 20 Q&A)

Q1: "Where is OpenGL or the GPU used in this engine?"

Answer: "Nowhere in the rendering pipeline. The GPU is completely bypassed. Every pixel of the line algorithms, circle symmetry, polygon rasterization, clipping, and rotation is calculated on the CPU in our custom C++ `SoftwareRasterizer` and written directly into an 800×300 `uint32_t` memory array in RAM."

Q2: "What is a Framebuffer in computer systems?"

Answer: "A Framebuffer is a linear, contiguous block of random access memory (RAM) allocated to hold the color values of every pixel on the display. In our system, it is an 800×300 flat 1D array of 32-bit ARGB words occupying 960 KB of memory."

Q3: "Why is floating-point arithmetic avoided in Bresenham's algorithm?"

Answer: "Floating-point division requires multiple CPU clock cycles and introduces cumulative rounding errors that can cause gaps or uneven steps. Bresenham reformulates the line equation into pure integer additions, subtractions, and bit shifts ($2 \times \text{err}$), making it exact and 10x faster."

Q4: "Why is the decision variable in Midpoint Circle initialized to $1 - r$ instead of $5/4 - r$?"

Answer: "Evaluating the midpoint $(1, r - 0.5)$ gives $F(1, r - 0.5) = \frac{5}{4} - r$. Because the radius r is always an integer, $\frac{5}{4} - r = 1 - r + 0.25$. Since 0.25 cannot change the integer sign comparison (value < 0), we replace $\frac{5}{4}$ with 1 to keep all calculations strictly integer."

Q5: "Why does the Midpoint Ellipse algorithm require two distinct regions?"

Answer: "An ellipse's slope changes continuously from 0 to $-\infty$. In Region 1, the slope magnitude is $|dy/dx| < 1$, making X the primary driver axis. In Region 2, $|dy/dx| \geq 1$, so stepping in X would leave holes; we must switch to stepping along the Y axis."

Q6: "How does the Parity Rule work in Scanline Polygon Filling?"

Answer: "For each scanline row, we find all intersections with polygon edges and sort them from left to right. Casting a horizontal ray toggles our fill state on odd-numbered intersections and disables it on even-numbered intersections, filling only the interior spans."

Q7: "Why did you use Inverse Mapping instead of Forward Mapping for Sprite Rotation?"

Answer: "Forward mapping rotates source pixels into destination coordinates, where rounding causes gaps and unpainted black holes. Inverse mapping iterates through destination pixels and samples backwards into the source using the inverse rotation matrix $R(-\theta)$, guaranteeing a continuous, solid sprite."

Q8: "Why does QImage wrap the framebuffer without memory copying?"

Answer: "We pass the raw pointer `reinterpret_cast<uchar*>(m_pixels.data())` directly to `QImage`'s constructor. This zero-copy wrapper enables 60 FPS presentation without wasting CPU bandwidth allocating or copying 960 KB memory buffers every frame."

Q9: "What is Coyote Time in game development?"

Answer: "Coyote Time provides a brief window (80 ms in our game) allowing the player to trigger a jump right after walking off a ledge, preventing player frustration from near-miss edge jumps."

Q10: "How does the RetroAudio engine generate sound without WAV files?"

Answer: "It synthesizes square waves $\text{sgn}(\sin(2\pi ft))$ mathematically in memory. By modulating frequency over time (e.g. 140 Hz to 420 Hz), it generates jump chirps and retro audio procedurally with zero external asset dependencies."